Robotic Systems-2026

LAB 4 REPORT

Euler-Lagrange equation of motion

Team members: Anas Khmais Hamrouni

1. Introduction

In this lab, we will derive the dynamics of the double-link pendulum using the Euler-Lagrange equation. Based on the obtained dynamics, we will simulate the free motion of the double link pendulum. The visualization function and sample implementation can be found here.

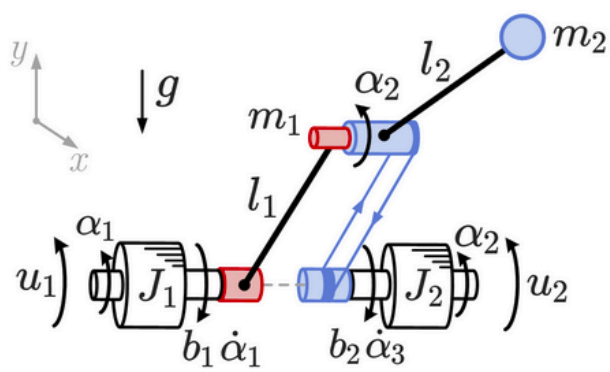
Fill in this report and **submit it to Moodle**.

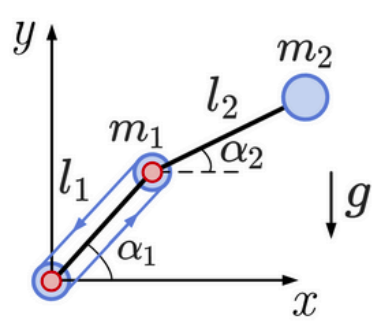
2. Simulation

Tasks:

2.1. Consider the double link pendulum rotating on the vertical plane. Derive the dynamics of the system using the Euler-Lagrange approach. Neglect the inertia on the motors.

$m_1 = 1$kg
$m_2 = 1$kg
$l_1 = 0.8$ m
$l_2 = 1.2$ m
$g = 9.8$ m/s^2





The point-mass positions are

$$\boldsymbol{r}_1 = \begin{bmatrix} l_1 \cos\alpha_1 \\ l_1 \sin\alpha_1 \end{bmatrix}, \qquad \boldsymbol{r}_2 = \begin{bmatrix} l_1 \cos\alpha_1 + l_2 \cos\alpha_2 \\ l_1 \sin\alpha_1 + l_2 \sin\alpha_2 \end{bmatrix}.$$

Velocities:

$$\dot{\boldsymbol{r}}_1 = \begin{bmatrix} -l_1\dot{\alpha}_1 \sin\alpha_1 \\ l_1\dot{\alpha}_1 \cos\alpha_1 \end{bmatrix}, \quad \dot{\boldsymbol{r}}_2 = \begin{bmatrix} -l_1\dot{\alpha}_1 \sin\alpha_1 - l_2\dot{\alpha}_2 \sin\alpha_2 \\ l_1\dot{\alpha}_1 \cos\alpha_1 + l_2\dot{\alpha}_2 \cos\alpha_2 \end{bmatrix}.$$

Hence,

$$\|\dot{\boldsymbol{r}}_1\|^2 = l_1^2\dot{\alpha}_1^2,$$
$$\|\dot{\boldsymbol{r}}_2\|^2 = l_1^2\dot{\alpha}_1^2 + l_2^2\dot{\alpha}_2^2 + 2l_1l_2\dot{\alpha}_1\dot{\alpha}_2\cos(\alpha_1 - \alpha_2).$$

Kinetic energy:

$$T = \frac{1}{2}m_1\|\dot{\boldsymbol{r}}_1\|^2 + \frac{1}{2}m_2\|\dot{\boldsymbol{r}}_2\|^2 = \frac{1}{2}(m_1+m_2)l_1^2\dot{\alpha}_1^2 + \frac{1}{2}m_2l_2^2\dot{\alpha}_2^2 + m_2l_1l_2\dot{\alpha}_1\dot{\alpha}_2\cos(\alpha_1 - \alpha_2).$$

Potential energy (with y positive upward):

$$V = m_1gy_1 + m_2gy_2 = (m_1+m_2)gl_1\sin\alpha_1 + m_2gl_2\sin\alpha_2.$$

Lagrangian:

$$\mathcal{L} = T - V.$$

Using

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\alpha}_i}\right) - \frac{\partial\mathcal{L}}{\partial\alpha_i} = 0, \qquad i = 1, 2,$$

the equations of motion are

$$(m_1+m_2)l_1^2\ddot{\alpha}_1 + m_2l_1l_2\cos(\alpha_1-\alpha_2)\ddot{\alpha}_2 + m_2l_1l_2\sin(\alpha_1-\alpha_2)\dot{\alpha}_2^2 + (m_1+m_2)gl_1\cos\alpha_1 = 0,$$
$$m_2l_1l_2\cos(\alpha_1-\alpha_2)\ddot{\alpha}_1 + m_2l_2^2\ddot{\alpha}_2 - m_2l_1l_2\sin(\alpha_1-\alpha_2)\dot{\alpha}_1^2 + m_2gl_2\cos\alpha_2 = 0.$$

Substituting the given numerical values:

$$1.28\,\ddot{\alpha}_1 + 0.96\cos(\alpha_1-\alpha_2)\,\ddot{\alpha}_2 + 0.96\sin(\alpha_1-\alpha_2)\,\dot{\alpha}_2^2 + 15.68\cos\alpha_1 = 0,$$
$$0.96\cos(\alpha_1-\alpha_2)\,\ddot{\alpha}_1 + 1.44\,\ddot{\alpha}_2 - 0.96\sin(\alpha_1-\alpha_2)\,\dot{\alpha}_1^2 + 11.76\cos\alpha_2 = 0.$$

2.2. Rewrite the dynamics derived from 2.1 in the form $M(q,\dot{q})\,\ddot{q} + C(q,\dot{q}) + g(q) = 0$, where $M(q,\dot{q})$ is inertia matrix, $C(q,\dot{q})$ is a matrix that corresponds to Coriolis forces, $g(q)$ is a matrix corresponding to gravitational forces.

Let $\Delta = \alpha_1 - \alpha_2$. Then

$$\boldsymbol{M}(\boldsymbol{q}) = \begin{bmatrix} (m_1+m_2)l_1^2 & m_2 l_1 l_2 \cos\Delta \\ m_2 l_1 l_2 \cos\Delta & m_2 l_2^2 \end{bmatrix},$$

$$\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}}) = \begin{bmatrix} 0 & m_2 l_1 l_2 \sin\Delta\,\dot{\alpha}_2 \\ -m_2 l_1 l_2 \sin\Delta\,\dot{\alpha}_1 & 0 \end{bmatrix},$$

$$\boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} (m_1+m_2) g l_1 \cos\alpha_1 \\ m_2 g l_2 \cos\alpha_2 \end{bmatrix}.$$

Numerically,

$$\boldsymbol{M}(\boldsymbol{q}) = \begin{bmatrix} 1.28 & 0.96\cos(\alpha_1-\alpha_2) \\ 0.96\cos(\alpha_1-\alpha_2) & 1.44 \end{bmatrix},$$

$$\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}}) = \begin{bmatrix} 0 & 0.96\sin(\alpha_1-\alpha_2)\,\dot{\alpha}_2 \\ -0.96\sin(\alpha_1-\alpha_2)\,\dot{\alpha}_1 & 0 \end{bmatrix},$$

$$\boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} 15.68\cos\alpha_1 \\ 11.76\cos\alpha_2 \end{bmatrix}.$$

Therefore,

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{0}.$$

2.3. Simulate the system dynamics derived from task 2.1 using Runge-Kutta or any other robust integrators (you may use odeint, but be aware that there can be some difficulty in implementation, which may consume your time). Visualize the motion using visualize.py.

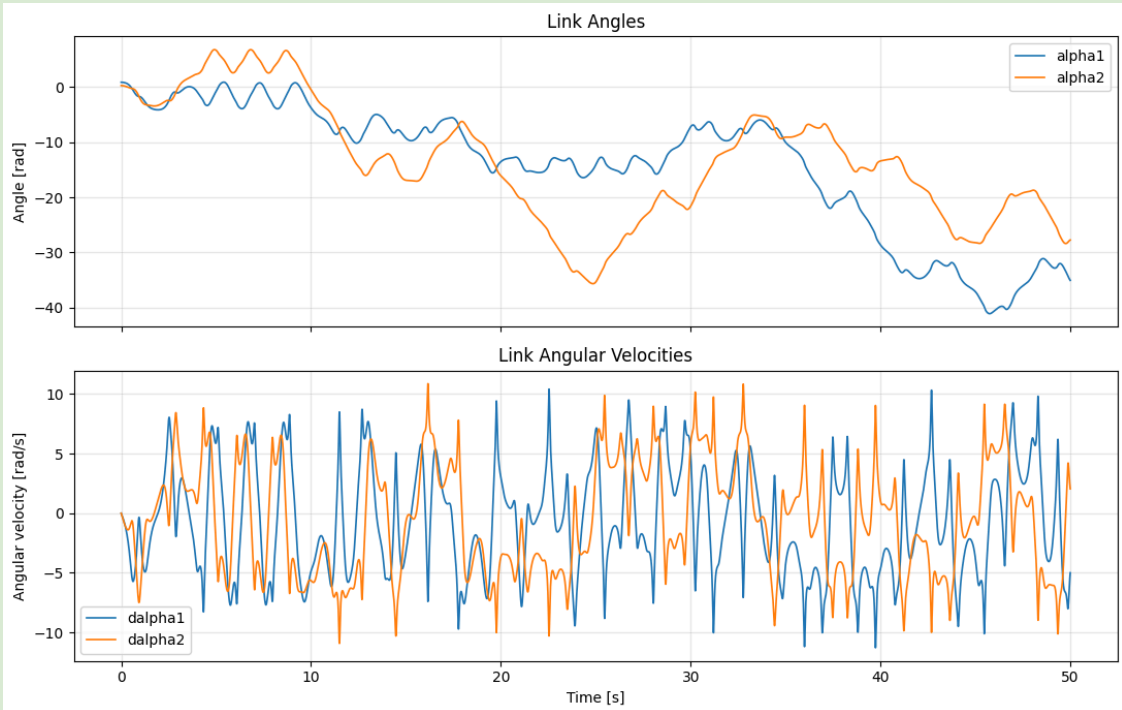
$t = 50$ s (total time of simulation) $dt = 0.01$ s (integration step)

Hint: First, define the state vector $x = [q, \dot{q}]^T$. From here you can get the derivative of state space vector as

$\dot{x} = [\dot{q}, \ddot{q}]^T = [\dot{q}, M^{-1}(q,\dot{q})(-C(q,\dot{q}) - g(q))]^T$.

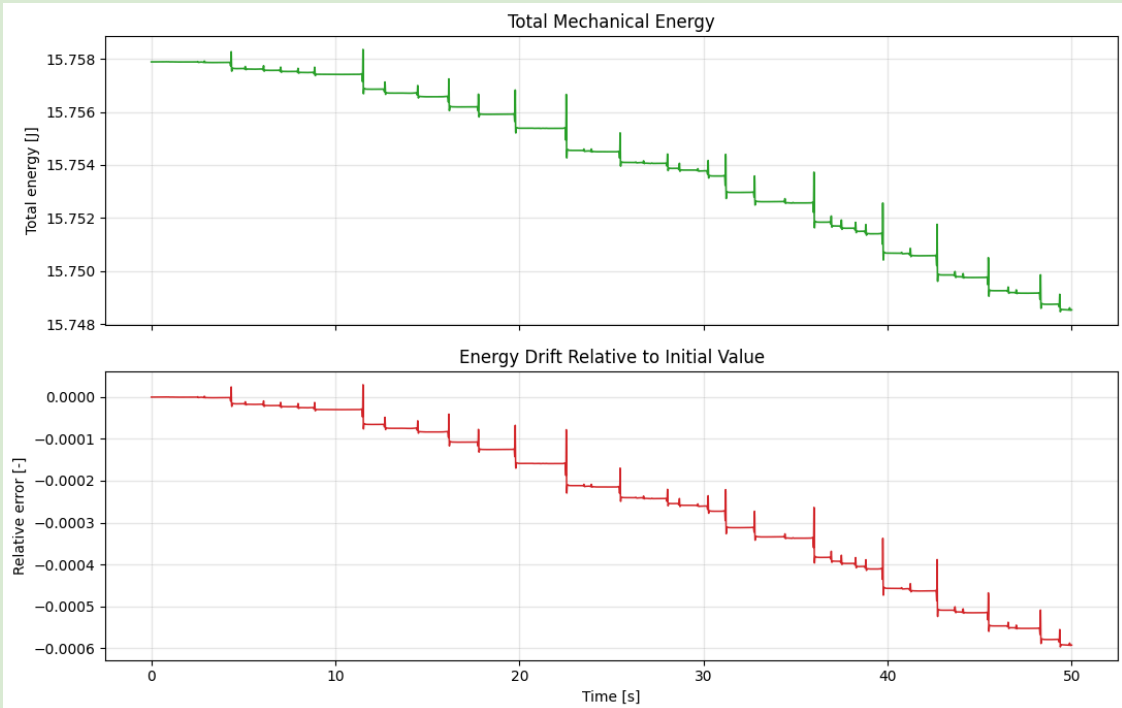
Simulation link
Code repository

2.4. Plot the angles and angular velocities of each link.



Plots for Task 2.4

2.5. Plot the full energy of the system.



Plots for Task 2.4

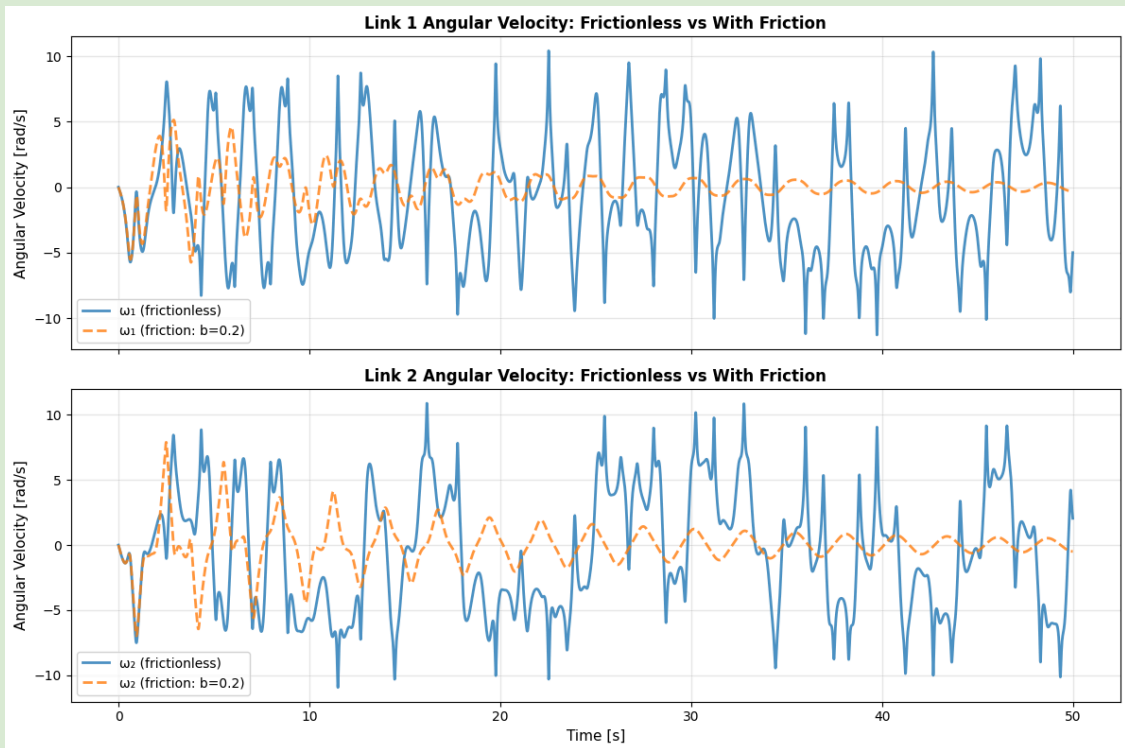
Explain the obtained results.

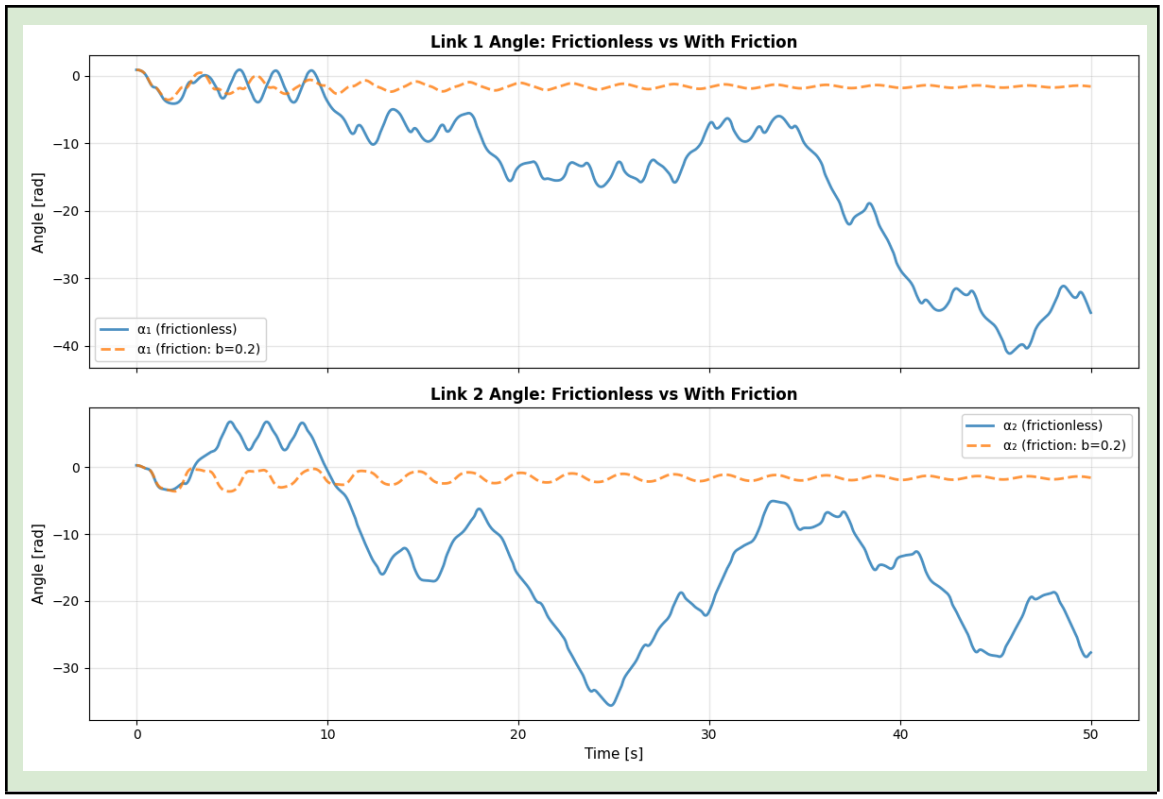
Answer(s):

1. The angle plots show coupled, nonlinear oscillations: neither link follows a pure sinusoid because both equations are coupled through `M(q)` and `C(q,dq)`.
2. The angular-velocity plots show phase-dependent energy exchange: when one link speeds up, the other often slows down, indicating transfer of kinetic energy between links.
3. The total mechanical energy is nearly conserved, as expected for a model without damping and actuator work. In this run, `E(0) = 15.757884 J` and the maximum relative drift is about `5.972e-04` (around `0.06%`).
4. The small drift is numerical (RK4 with finite `dt=0.01`). Using a smaller `dt` should reduce this drift further.
5. The animation is consistent with the plots: faster motion appears near lower potential-energy configurations, while motion slows near turning points.

2.6. Consider friction forces on each link $M(q,\dot{q})\,\ddot{q} + C(q,\dot{q}) + g(q) + Q_f = 0$, where $Q_f = [b_1\dot{\theta}_1,\ b_2\dot{\theta}_2]$ ($b_1 = b_2 = 0.2$ are friction coeficients). Plot the angles and angular velocities of each link and compare the results with the case without friction.

Plots for Task 2.5





Explain the obtained results.

Answer(s):
In the frictionless case, the two links keep exchanging energy through the nonlinear coupling, so the angles and angular velocities remain oscillatory over the whole simulation, while the total mechanical energy stays essentially constant. In the friction case, the velocity-proportional friction terms remove energy from the system, so both links gradually lose amplitude and the angular velocities decay toward zero.